

SEMESTER-IV

URA401: DATA SCIENCE				
	L	T	P	Cr
	2	1	2	3.5
Course Objective: To introduce the concepts and tools of data science. By the end of the course, students will have gained proficiency in scientific programming, data manipulation, feature engineering, data visualization, and statistical analysis.				
Syllabus				
Introduction to Data Science				
Introduction to Data Science, Data science lifecycle, Data sourcing techniques				
Statistics for Data Science				
Measures of central tendency, Measures of dispersion (variance, standard deviation), Probability distributions in data analysis, Hypothesis testing (Z-test, t-test, chi-square test), Confidence intervals, Estimation techniques, Maximum Likelihood Estimation (MLE)				
Exploratory Data Analysis (EDA) and Data Visualization				
Data summarization techniques, Handling missing and inconsistent data, Outlier detection and treatment, Correlation and covariance analysis, Identifying data patterns and trends, Distribution Analysis, Basic plotting using Matplotlib, Line plots, scatter plots, and bar charts, Histograms and density plots, Box plots and violin plots, Subplots and multiple plots in a single figure, Heatmaps and contour plots, Introduction to 3D plotting				
Feature Engineering & Dimensionality Reduction				
Introduction to feature engineering, Role of features in machine learning models, Feature scaling and normalization, Feature extraction techniques, Principal Component Analysis (PCA), Dimensionality reduction concepts and applications				
Introduction to Machine Learning				
Overview of machine learning, Supervised vs unsupervised learning, Linear regression (conceptual and practical), Logistic regression (classification basics), K-Means clustering, Model evaluation metrics (accuracy, precision, recall, F1-score)				
Laboratory Work				
1. Writing Python programs for understanding basic syntax, variables, operators, control flow statements in Python. 2. Writing Python programs for working with various data structures such as lists, tuples, sets, and dictionaries. 3. Writing Python programs to use NumPy arrays to perform vectorization and broadcasting on arrays and apply various array operations such as sorting and searching. 4. Writing Python programs to read and manipulate data from various sources using Pandas data structures such as Series and DataFrames and apply various operations such as handling missing data, merging and joining, renaming columns etc..				

Approved in 113th meeting of the Senate held on September 7, 2024, respectively. Further, revised in 1st meeting of the Academic Council held on September 11, 2025.

5. Writing Python programs to use Matplotlib to create various types of plots including line plots, scatter plots, bar plots, and histograms.
6. Writing Python programs to apply feature engineering techniques to extract useful information from data.
7. Writing Python programs to apply PCA to reduce the dimensionality of a dataset and extract important features.

Course Learning Objectives (CLO)

The students will be able to:

1. Explain the data science lifecycle.
2. Perform statistical analysis including descriptive statistics, hypothesis testing, confidence intervals, and parameter estimation.
3. Conduct exploratory data analysis and create effective visualizations to identify patterns and insights from data.
4. Implement basic machine learning techniques, feature engineering, and dimensionality reduction to analyze real-world datasets.

Text Books

3. Eric Matthes, —*Python Crash Course: A Hands-On, Project-Based Introduction to Programming*l, 2019, 2nd Edition, Packt Publishing.
4. Müller, A. C., and Guido, S., *Introduction to machine learning with Python: a guide for data scientists*, O'Reilly Media, Inc. (2016).

Reference Books

1. Vijayvargia Abhishek, *Machine Learning with Python*, BPB Publication (2018).
2. *Data Science Using Python and R*, by Chantal D. Larose, Daniel T. Larose, Wiley (2019).

Evaluation Scheme:

Sr.No.	Evaluation Elements	Weightage (%)
1.	MST	25
2.	EST	40
3.	Sessionals (May include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	35

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